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TEER

Mitral Regurgitation, MR 2008 Mitral Valve Transcatheter Edge-to-Edge Repair, M-TEER MR

Mitral Regurgitation, MR 10% 75 50% MR

MitraClip Alfieri 2003 Dr. Feldman EVEREST I EVEREST II 12

M-TEER guideline M-TEER review Mainz Primary Mitral Regurgitation, PMR Secondary Mitral Regurgitation, SMR MR

1.1 Background

Mitral Regurgitation, MR 10% 75 MR³. MR Heart Failure, HF⁴ MR

2003 Mitral Valve Transcatheter Edge-to-Edge Repair, M-TEER 2008 CE European Conformity 2013 2019 FDA MR Primary MR, PMR MR Secondary MR, SMR

1

2

³; and

4

5

謝國祥醫師於2023年2-8月間，
MR. 謝國祥醫師

Only 5.6% of patients who underwent isolated MV surgery in the Society of Thoracic Surgeons (STS) database from 2008 to 2012 had an STS-predicted mortality rate of >12% (response to a query of the STS database conducted in December 2012 of all isolated MV surgeries between 2008 and 2012).⁶

⁶
STS score > 12% trials high-risk patient definition.

1.3 MR and TR evaluation

1.3.1 Clinical evaluation

Heart Failure, HF New York Heart Association, NYHA¹¹

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Heart Failure, HF Kansas City Cardiomyopathy Questionnaire (KCCQ) (Table ??). (Table ??)

(right heart catheterization) (coronary angiography) Coronary Revascularisation¹²

12; and

Coronary Computed Tomography Angiography

Tricuspid Regurgitation, TR Valvular Heart Disease, VHD

Table 1.1: Kansas City Cardiomyopathy Questionnaire (KCCQ)

Item	Scale
Physical limitation	0 = Severely limited to all activities; 100 = No limitation at all
Quality of life	0 = Very poor; 100 = Excellent
Angina	0 = No angina; 100 = Severe angina
Shortness of breath	0 = No shortness of breath; 100 = Severe shortness of breath
Swelling	0 = No swelling; 100 = Severe swelling
Confidence in doctor	0 = No confidence; 100 = Complete confidence
KCCQ summary score	0 = Worst possible health; 100 = Best possible health

1.3.4 Palliative care and LVAD/heart transplantation

對於嚴重心臟衰竭 (class III or IV) 且對藥物治療無反應者，可考慮 Palliative Care 或心臟移植。心臟衰竭 (Heart Failure, HF) 合併二尖瓣反流 (Mitral Regurgitation, MR) 患者，若經藥物治療後症狀仍無改善，可考慮 Left Ventricular Assist Device (LVAD) 或心臟移植 (Heart Transplantation, HTx)。

1.3.5 Echocardiographic evaluation- etiology

根據 Carpentier's classification (Figure ??). 根據 MR 的病因可分為 Primary MR¹⁴

¹⁴; and

Primary Mitral Regurgitation, PMR。PMR 的病因包括 chordae 異常 (Carpentier II) 或 leaflet 運動受限 (Carpentier IIIa)。PMR 的診斷可參考 Figure ??。

Secondary Mitral Regurgitation, SMR 的病因包括 annular dilatation、leaflet 異常 (Carpentier I)、LV remodeling (LV dilatation) 或 leaflet 異常 (Carpentier IIb)。SMR 的診斷可參考 Figure ??。

Secondary Mitral Regurgitation, SMR 可分為 ischemic 或 non-ischemic。atrial 異常¹⁵

¹⁵

- SMR (ischemic secondary MR) 的病因包括 Left Ventricle (LV) 擴大導致 leaflet 閉合不全 (Hypokinesia) 或 asymmetrical MV tethering。
- SMR (non-ischemic secondary MR) 的病因包括 Dilated Cardiomyopathy 導致 symmetrical MV tethering。
- SMR (atrial functional secondary MR) 的病因包括 Atrial Fibrillation (AF)。

1.3.6 What is tethering?

Tethering 是什麼？

Mitral Valve Tethering
 Mitral Regurgitation
 Secondary Mitral Regurgitation
 16

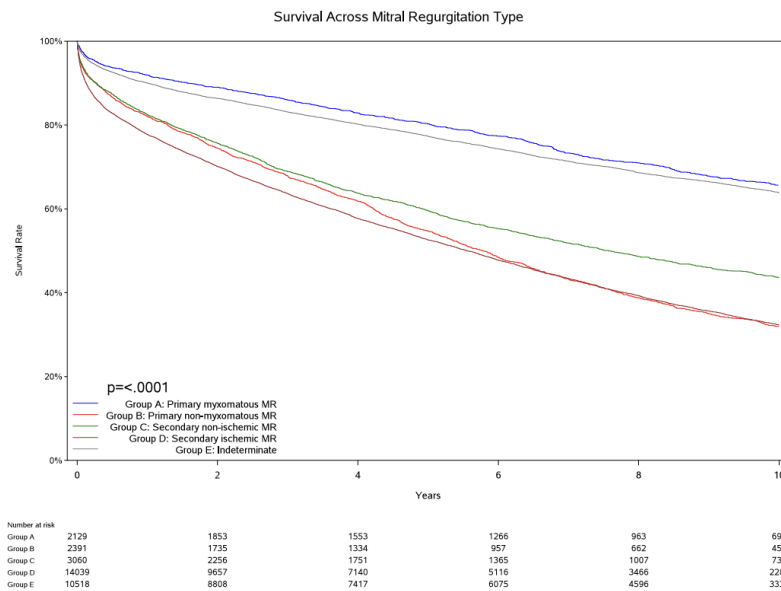


Figure 1.2: Unadjusted Survival Curves for Different Etiologies of Mild MR. Unadjusted survival of patients with mild MR stratified by etiology. Abbreviations: MR, mitral regurgitation.^a

^a [

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1.3.7 Echocardiographic evaluation- repair feasibility

MPR (multi-plane reconstruction)
 Measurement for leaflet grasping (mandatory in Mainz):

- Mitral annular diameter: AP and intercommissural planes: > 38-40 mm for XTW or XT
- Mitral valve area: > 4 cm²
- Grasping zone leaflet mobile length

1.3.8 The Carpentier classification

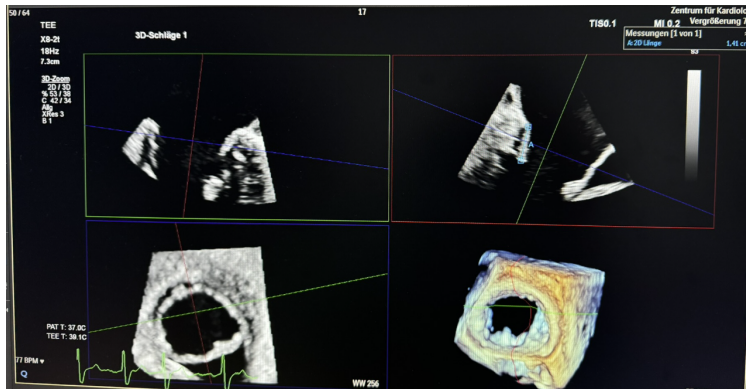


Figure 1.3: MPR, multiplanar reconstruction, □□□□□□□□□□□□□□□□□□
□□

Table 1.3: Carpentier classification of MR/TR

Class	Description
Class I	MR secondary to aortic regurgitation or aortic coaptation delay. MR is usually mild to moderate.
Class II	MR due to prolapse or flail leaflet. Primary Mitral Regurgitation, PMR
Class III	- Class IIIa: MR due to restrictive cardiomyopathy or left ventricular hypertrophy. - Class IIIb: MR due to ventricular remodeling (e.g., DCM-related ventricular dilatation) or asymmetrical restriction (e.g., ischemic cardiomyopathy or posterior leaflet tethering).

1.3.9 Echocardiographic evaluation- severity

Qualitative Mitral Regurgitation, MR
 Qualitative Semi-Quantitative Quantitative
 1 MR

- Vena Contracta width ≥ 7 mm
- Regurgitant Fraction $\geq 50\%$
- Effective Regurgitant Orifice Area, EROA ≥ 40 mm²
- Regurgitant Volume, RVol ≥ 60 ml

DCMP, EROA, RVol
 PISA Proximal Isovelocity Surface Area
 Effective Regurgitant Orifice Area, EROA Regurgitant
 Volume, RVol Secondary Mitral Regurgitation,
 SMR VCW and regurgitant fraction¹⁷

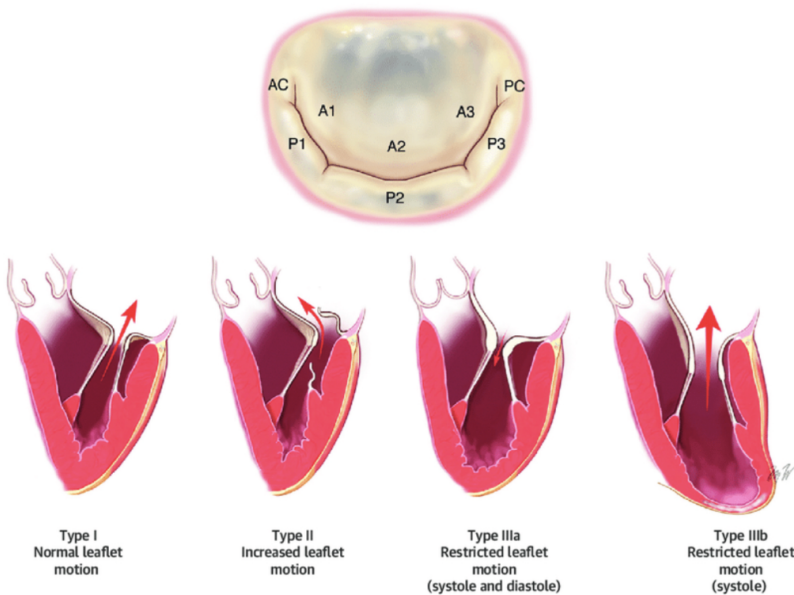


Figure 1.4: Carpentier classification. Carpentier classification Mitral Regurgitation, MR; and

^a [

Table 1.4: Mainz Heart Valve Center 經驗

項目	標準
Vena contracta width	≤ 7 mm
E-velocity	≥ 1.5 m/sec
VTI of MR	≤ 1.4 cm
Pulmonary vein flow reversal	MR 嚴重

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1.3.10 When should we NOT proceed with M-TEER?

何時不應進行 M-TEER? Mainz 經驗: severe mitral stenosis, 2-6 mmHg

- Annular diameter (anterior-posterior) < 30 mm.
- Severe valve calcification.
- Resting mitral inflow pressure gradient ≥ 5 mmHg.

1.4 Special note on patients with very low LVEF

低 LVEF 患者 ERO 和 regurgitant volume (RV) 評估
 LV contractility 評估 ERO 和 RV
 2024-11-12 case 19 (Figure ??) MR color mapping jet
 LA 40% area, ERO 0.1 cm², RV 12 ml. vena contracta width 13 mm, pulmonary vein flow reversal (severe MR).

1.5 MV anatomy

Indentation lines: posterior leaflet indentation
 M-TEER leaflet tension M-TEER MR
 Findings suggestive of only mild MR:

- prominent A

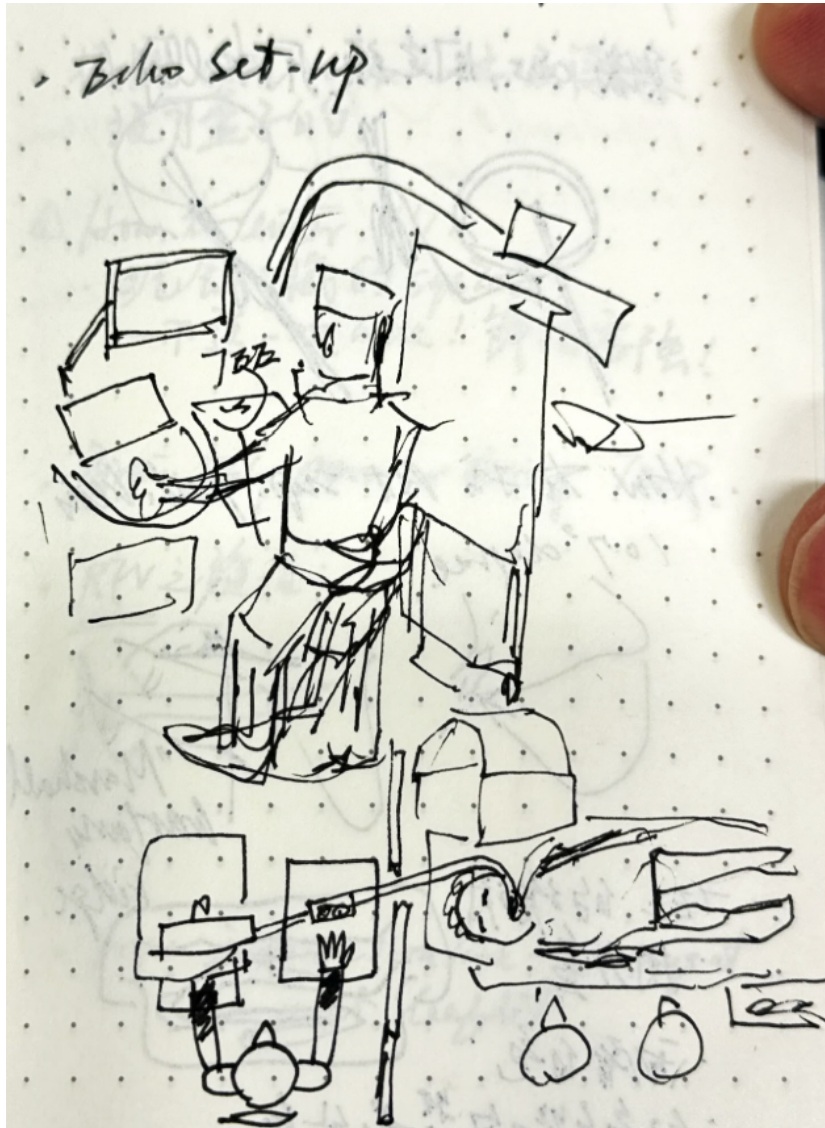


Figure 1.7: TEE operator TEE operator

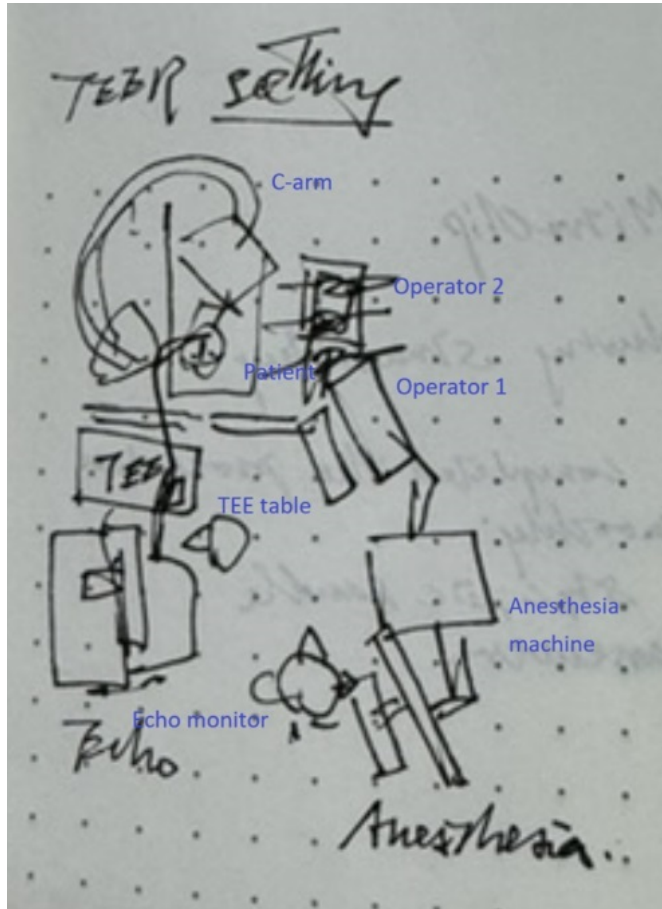


Figure 1.8: Operator 1 □□ operator 2
 □□□□□□□□□□□□□□□□□□□□
 □ Philips □□□□□□□□□□□□□□□□
 □ single arm, □□□□□□□□□□□□□□
 interventional TEE operator. □□□□□□:
 □□□□□□□□□□□□□□□□□□□□

- 2D color mapping with X-plane, to further confirm the pathology relation to jet.
- 3D echo, with MPR, to evaluation if there is any jet missed from the previous evaluation sequence.



Figure 1.9: TEER echo machine

1.7 Treatment algorithm

(A) Primary Mitral Regurgitation, MR (B) Secondary MR Figure ??.

- COAPT MitraClip Cardiovascular Outcomes Assessment of the MitraClip Percutaneous Therapy for Heart Failure Patients With Functional Mitral Regurgitation

- GDMT方方方方方方方方方方Guideline-Directed Medical Therapy方方
- HTx方方方方方方Heart Transplantation方方
- LVAD方方方方方方方方方方Left Ventricular Assist Device方方
- M-TEER方方方方方方方方方方方方方方Mitral Valve Transcatheter Edge-to-Edge Repair方方
- SMVR方方方方方方方方方方方方方方Surgical Mitral Valve Repair or Replacement方方

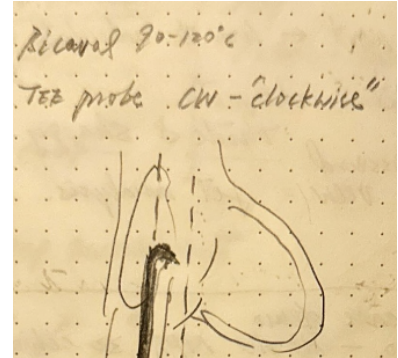


Figure 1.10: Bi-caval view 方方方方方方方方方方 trans-septal puncture 方方方方方方 view.

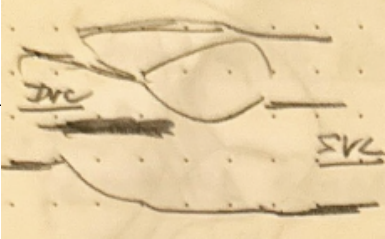


Figure 1.12: Bi-caval view echocardiogram showing a trans-septal puncture. The image displays the right and left atria with a needle visible through the septum. Handwritten labels 'DVC' and 'SVL' are present on the image.

1.8 TEER repair systems

Table 1.5: MitraClip/TriClip vs PASCAL

	MitraClip/TriClip	PASCAL
Company	Abbott	Edwards Lifesciences
Device	MitraClip/TriClip	PASCAL
Spacer	Spacer	Spacer
Spacer (Spacer)		
Spacer	Spacer	Spacer
Spacer	Spacer	Spacer
Spacer	Spacer NT, XT	Ace
Spacer	Spacer	PASCAL Precision

1.8.1 MitraClip

MitraClip is a transcatheter aortic valve repair device.

1.8.2 PASCAL

M-TEER and T-TEER.

First reported the use to PASCAL to treat severe TR. Reported PASCAL use as compassionate therapy for patients with severe TR.

If annular diameter > 40 mm, always consider MitraClip XT or XTW first. If annular diameter < 35 mm, the only options are MitraClip NT or NTW.

1.9 TEER procedure steps

1.9.1 Preparation of MitraClip/TriClip

1.9.2 Steerable guiding catheter (SGC) placement

heparin: SGC

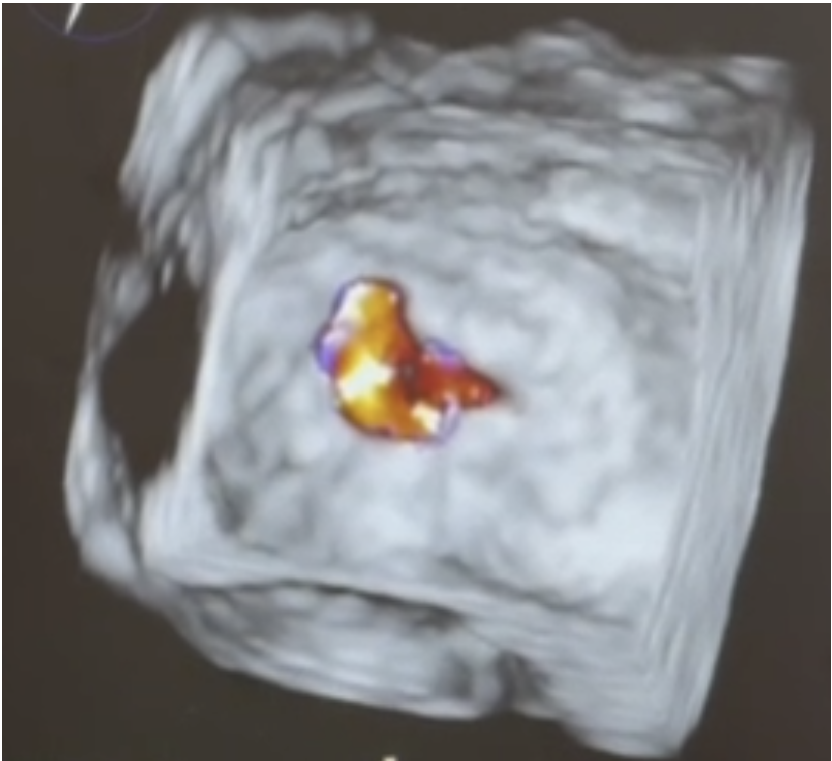


Figure 1.13: Jet analysis TEER Mainz X-plane, 3D, MPR jet analysis. jet jet leaflet pathology

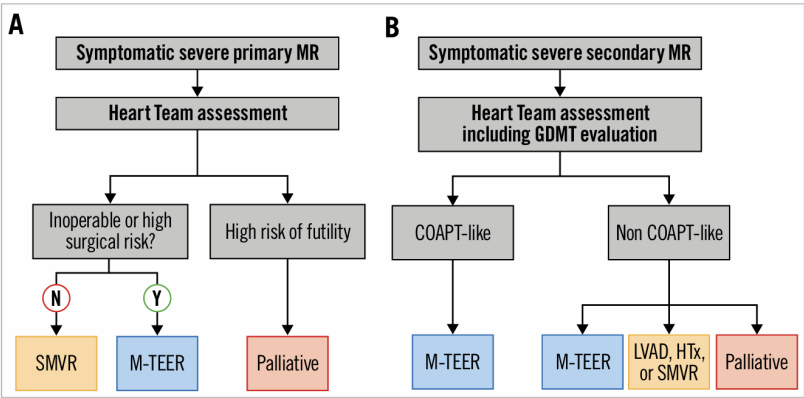


Figure 1.14: Patient stratification and simplified guideline recommendation. Therapeutic strategies for patients with (A) symptomatic severe primary mitral regurgitation (MR) or (B) symptomatic severe secondary MR.^{a]}

^{a]} [

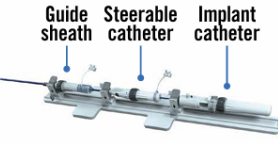
	MitraClip (4 th -generation)	PASCAL Precision (2 nd -generation)
Delivery catheter		
Available implants	 NT NTW XT XTW	 P10 ACE
Device material	Rigid arms of cobalt-chromium alloy	Flexible arms of nitinol
Central spacer	No	Yes
Optional independent grasping	Yes	Yes
Closure mechanism	Active (locking element)	Passive (nitinol shape memory)
Number of working catheters	2	3
Orientation of hooks/friction elements	Longitudinal	Horizontal
Continuous LA pressure	Yes	Yes
Overall system stability	High	Improved with PASCAL Precision

Figure 1.15: Mitral Valve, MV TEER MitraClip Abbott PASCAL Edwards Lifesciences Table ??

1.9.3 Use of fluoroscopy

Fluoroscopy interventional TEE TEER RAO cranial or caudal, clip fluoroscopy ventricle clocking²⁰

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fluoroscopy clip ventricle fluoroscopy clocking

1.9.4 Transseptal puncture

1.9.5 Before the era of interventional TEE

1.9.6 In the era of interventional TEE

(Brockenbrough needle) (Mullin sheath) Brockenbrough 5 6 4 6 SVC TEE TEE

The needle and the sheath are held in the fingers to prevent inadvertent advancement of the needle, and the position of the Brockenbrough needle arrow is at 5 or 6 o'clock, according to different procedures (Figure 4). To target a more posterior puncture location, the arrow should be kept at the 6 o'clock position. The needle and the sheath are pulled back caudally until they are visualized

in the inferior portion of the SVC. Tenting should be monitored by echocardiography (bicaval view) in the lower part of the SVC before the needle falls into the fossa. Usually, as the needle tents the superior rim of the fossa, supraventricular extrasystolic beats are generated. During this phase, the TEE operator should avoid moving the image from the membranous portion of the septum, and the interventionalist should try to move in the direction of the image.

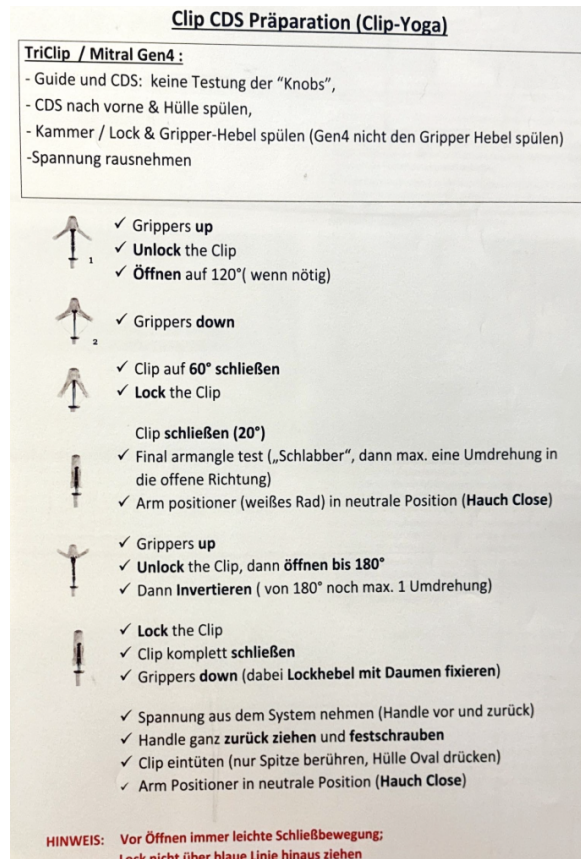


Figure 1.16: □□□ Mainz hybrid OR □□□ clip □□□□□

AP projection 1 AP
2

00000000000000TEE00000000000000000000000000“0
0”0tenting00 5A0000TEE 000000SAX00000000“00”000000000
000000000000“00”00000000 5B00

transseptal operator
operator
operator
——“”

“”

MitraClip Abbott Vascular TEE 4-chamber view (tenting point to mitral annulus distance 5C (Carpentier classification II) 4 (Carpentier classification III) 3.5

MitraClip Cardioband Edwards Lifesciences LAAO (TEE 5D

TEE SAX Fluoroscopy AP projection SVC Waterston

RF NRG RF Transseptal Kit Baylis Medical Company, Inc.

tenting TEE 100 U/kg ACT 250 300

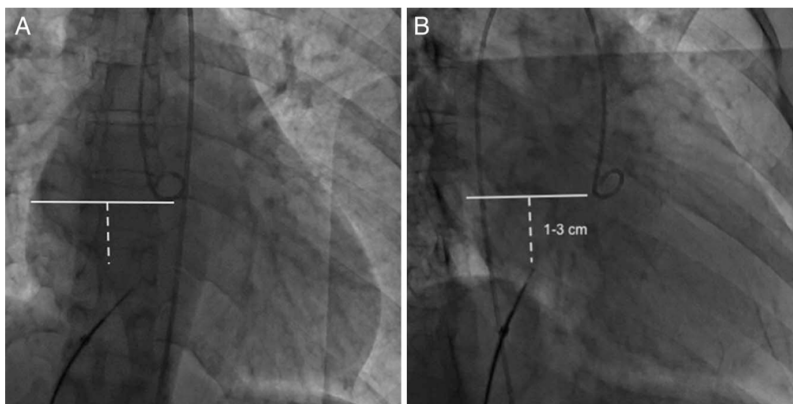


Figure 1.17: The fluoroscopy-guided transseptal puncture landmarks. (A) In the anteroposterior projection, the puncture site should be halfway from the distance of the catheter and the right border of the left atrium and one vertebral body height lower to the aortic valve plane (seen by inserting a pig tail catheter in the non-coronary aortic sinus). (B) In the 40° right anterior oblique projection, confirm if the needle is positioned neither close to the aortic root (too anterior) nor close to the free wall of the right atrium (too posterior). The puncture site should be 1–3 cm below the midpoint of a line connecting the posterior wall of the aorta to the back wall of the right atrium.

1.9.7 Insertion of steerable guide catheter (SGC)

1.9.8 Straddling

Straddling □□□□ MitraClip □□□□□□□□□□ SGC □ CDS □□□□□□□□□□□□
 □□□□□□□□ device □□□□ PASCAL device □□□□□□□□□□□□□□□□□□□□
 □□ SGC □ CDS □□□□□□

1.9.9 Adjustment for trajectory

□□ TEER □□□□□□□□□□□□ operator □□□□ MitraClip, TriClip, or
 PASCAL □□□□□□□□□□ knob (□□) □□□□□□□□□□□□□□□□□□□□□□□□
 Mainz □□□□□□□□□□□□□□□□□□□□

□ Trajectory adjustment □□ device enter to target valve □□□□□□
 □□□□□□ entering to LV or RV □ trajectory □ perpendicular to valve
 orifice. □□□□ central position, posterior trajectory, □□□ flex guide
 sheath (PASCAL) □□□ counter-clockwise rotation of guide sheath
 (PASCAL). □□□ Figure ??□□ Figure ??.

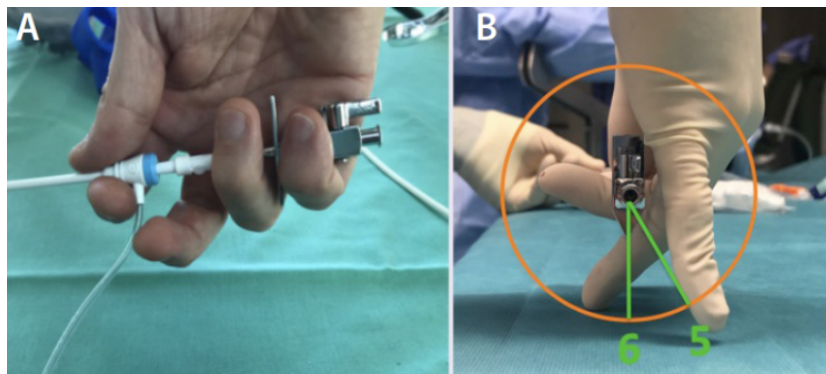


Figure 1.18: Correct handling (A) and orientation (B) of the Mullins sheath and Brockenbrough needle during the pullback maneuvers. Figure from <https://citoday.com/articles/2019-may-june/transseptal-puncture-a-step-by-step-procedural-guide>

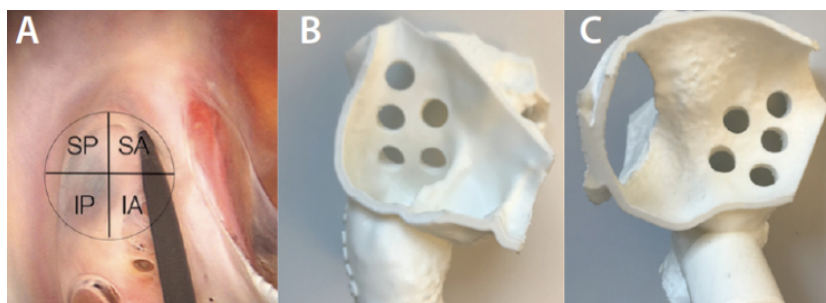


Figure 1.19: Fossa ovalis anatomy from the RA view and its schematic division into four parts (A). Three-dimensional models showing the IS from the right side (B) and the left side (C). IA, inferior-anterior; IP, inferior-posterior; SA, superior-anterior; SP, superior-posterior. Figure from <https://citoday.com/articles/2019-may-june/transseptal-puncture-a-step-by-step-procedural-guide>

1.9.10 Check gripper and adjustment for orientation

□□□□□□□□ clip/clasp □□□□□□□□ mitral leaflet □□□□□□□□□□□□□□
 □ clip/clasp □□□□□ leaflet coaptation line □□□□□ (Figure ??)□Figure
 ??□□□□□□□□ A2/P2 grasping orientation.



Figure 1.22: □□□□□□□□□□ Clasp (PASCAL) tip □□□□□- posterior trajectory, □□ tip □□□□□ mitral valve orifice □□ □- central position. □□□□□□□□□□□□□□ □□□□□ posterior trajectory □□□□□ central trajectory.

1.9.11 Leaflet grasping

TEE lower-down gripper leaflet insertion clip/clasp (Figure ??) color mapping, post-clipping echo evaluation (Figure ??).

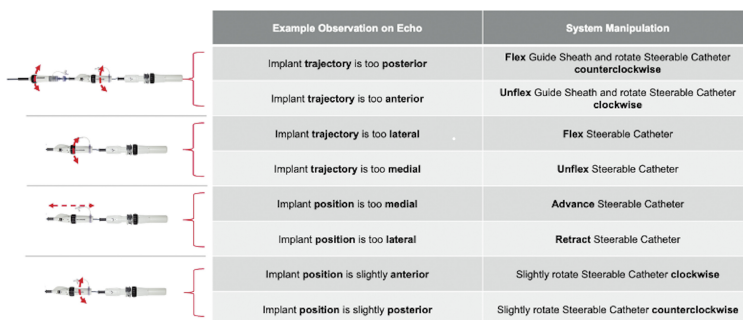


Figure 1.23: α trajectory and position α clip/clasp α α



Figure 1.24: TEER system components: MitraClip, steerable guide catheter (SGC), catheter delivery system (CDS), clip and orientation device

1.9.12 Final deploy

Table 1.6: MitraClip/TriClip deploy sequence (Figure 1.25)

Step	Action
1	Locker lock (MitraClip/TriClip)
2	Clip arm positioner (MitraClip/TriClip) rotate counter-clockwise (CCW), counter-clockwise rotation (MitraClip/TriClip) lock (MitraClip/TriClip) rotate clockwise (CW), clockwise rotation (MitraClip/TriClip)
3	Rotate CCW (MitraClip/TriClip) clip arm positioner (MitraClip/TriClip) neutral position (MitraClip/TriClip)
4	Deploy (MitraClip/TriClip)
5	Clip arm positioner (MitraClip/TriClip)
6	CDS (MitraClip/TriClip)
7	Cone (MitraClip/TriClip)
8	Deploy (MitraClip/TriClip)
9	Gripper (MitraClip/TriClip)
10	CDS (MitraClip/TriClip) SGC (MitraClip/TriClip)
11	Suture (MitraClip/TriClip)

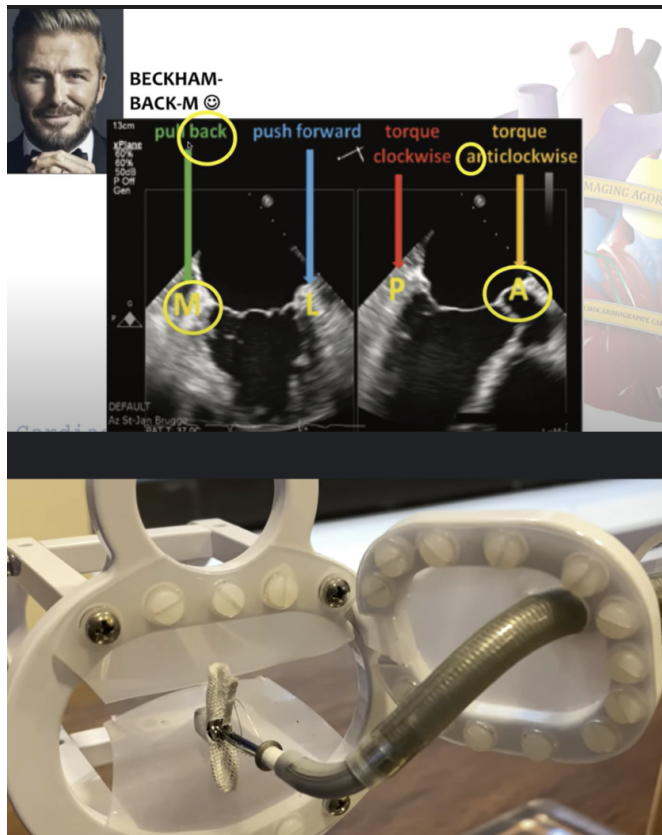


Figure 1.25: MitraClip M-L knob, TEE monitor panel, intercommissural plane clip mitral valve

- jet color mapping area
- mitral and tricuspid inflow velocity and mean pressure gradient (M-TEER mitral inflow mean PG should < 5 mmHg)
- mitral valve area: should $> 1.5 \text{ cm}^2$
- tricuspid valve area: should $> 2.5 \text{ cm}^2$

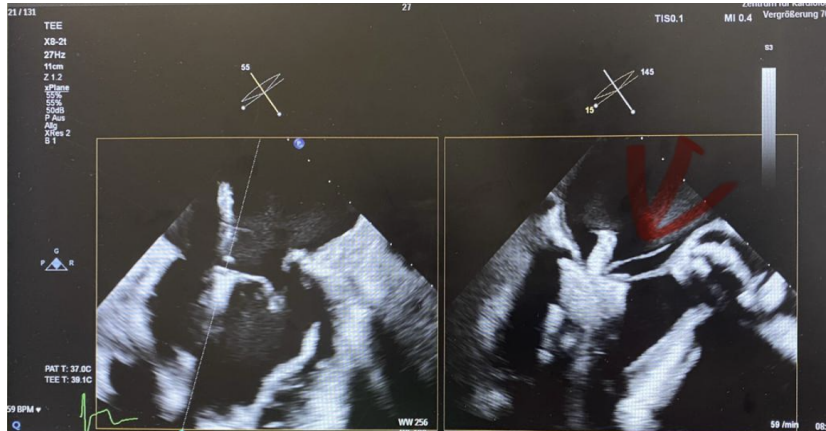


Figure 1.27: 醫師 (grasping) 醫師
醫師醫師醫師醫師醫師醫師醫師醫師醫師醫師:
leaflet becomes immobile. 醫師醫師
醫師 congenital anomaly: congenital
cord extending from anterior leaflet to
inter-atrial septum.

1.13 Reporting standards- TEER

1.13.1 Mainz

Größe: 150 cm Gewicht: 47 kg

Station: M2PAUF

HK-Datum: 01.07.2024

1. Untersucher: Dr. med. T. Ruf Steriles Assistenzpers.: Wolf, Dominik
Univ.-Prof. Dr. Dr. med. Ph. Lurz

2. Untersucher: Unster. Assistenzpers.: Lengwens, Julia

Herzchirurg: Dr. med. T. Ruf

Intervention: Dr. med. T. Ruf

Punkteur: Dr. med. T. Ruf

Diagnose: **Primäre Mitralklappeninsuffizienz III. Grades (Flail des PMLs, Carpentier 2)**
- **interventionelle Edge-to-Edge Mitralklappenrekonstruktion, 1x**
MitraClip XTW

Weitere Therapie/ Aspirin (1 x 100mg) für 6 Monate und Clopidogrel (1 x 75mg) für 3 Monate.
Procedere: Druckverband bis 16:00 Uhr. Hautnaht am nächsten Tag entfernen.
Endokarditisprophylaxe bei entsprechender Exposition für 6 Monate empfohlen.
UKG im Verlauf zum Ausschluss PE und MI Reduktion. Vorstellung in der
Klappen-Ambulanz in 1-3 und 12 Monaten.

Indikation: Die rechtfertigende Indikation zur Durchführung der Untersuchung wurde von
einem Facharzt bestätigt. Ein Team-Time-Out ist vor Beginn der Untersuchung
erfolgt.

Intervention: Mitralklappeninsuffizienz
Heart Team Entscheidung zur interventionellen Therapie bei log ES 21,69 %
und Alter 86,0 Jahre. Punktion über Vena femoralis rechts (25F). Reparatur der
Mitralklappe durch Edge-to-Edge. Eingriff an der Mitralklappe mittels eines
MitraClips 1. Problemlose transeptale Punktion mit Brockenbrough-Nadel via
SL1-Schleuse. Implantation eines MitraClip XTW in das laterale Segment 2 mit
jeweils guter Segelinserion. Abschließend leichtgradige Mitralsuffizienz
(Reduktion der MI von 3 auf 1), MV Gradient nachher 1 mmHg. latrogener
Septumdefekt mit links-rechts Shunt. Kein Perikarderguss.

Hämodynamik:

Erbrachte 5-35a.41 - Mitralklappensegelplastik, transvenös
Leistung: 3-992 - Intraoperative Anwendung der Verfahren
1-274.30 - Mit Nadel
8-700.0 - Durch oropharyngealen Tubus
3-991 - Computergestützte Bilddatenanalyse mit 4D-Auswertung

Komplikationen: Keine Komplikationen

Figure 1.32:

Univ.-Prof. Dr. Dr. med. Ph. Lurz Univ.-Prof. Dr. Dr. med. Ph. Lurz
Direktor der Klinik

1.13.2 Example of a TEER report

Mainz, Germany formal report

2025-03-07

MR, severe, primary,
Carpentier 2, with P1 flail, and pulmonary vein flow reversal,
s/p MitraClip (NT x1, NTW x1 to zone 1 and zone 2), success-

國立臺灣大學醫學院附設醫院新竹盛大分院新竹醫院
National Taiwan University Hospital Hsin-Chu Hospital

病歷號: H274077
姓 名: 何戴素麗

生 日: 19410101
檢查單號: 74287513156

年齡: 85
性別: F
檢查日期: 20250307

心導管檢查報告

冠狀動脈血管攝影
適應症Indication :
Mitral valve disease
() Primary PCI

術式Procedure:
(V) Other
M-TEER (MitraClip)

遠端Vascular access :
RFV
RFV
Sheath: 8Fr
Sheath: 25Fr
notes:
notes:

麻醉Anesthesia :
(V)Local 2% xylocaine,dose
JL3
JL3
8 ml

SHNTAX Score :
NA

血行動力學
Hemodynamics record
LRA 110 / 70 mmHg,mean ,Sat 02%
Final 123 / 72 mmHg,mean ,Sat 02%
Result (V)Success ()Fail
Complication: ()Yes (V)No
Type:

Figure 1.33: 何戴素麗

MR, severe, primary, Carpentier 2, with P1 flail, and pulmonary vein flow reversal, s/p MitraClip (NT xl, NTW xl to zone 1 and zone 2), successful,

Management:
Eliquis 2.5 mg po bid until 1 day before operation.
Plan to continue in the coming morning.

Risks of open heart surgery for MR:
. Operative mortality: 10.8%
. Morbidity and Mortality: 24.6%
. Stroke: 3.85%
. Renal failure: 5.07%
(by ascdriskcalc.research.sts.org, accessed 2025-03-09)

Procedure steps:
. Time-out in standard manner, with operation consent obtained. Self-paid 1 million NTD.
. A-line at Lt high-radial artery by ultrasound-guided puncture (by anesthesiologist).
. Ultrasound-guided puncture to RFV, with pre-set of ProGlide.
. Transeptal puncture with SL0 under TEE guidance, smoothly.
. Serial dilation and placement of 25 Fr SGC.
. Placed NT to A1/P1 and NTW to A2/P2, with good leaflet insertion.
. LAP improved from peak 59 mmHg (giant V wave) to 18 mmHg.
. Post M-TEER mitral inflow gradient: 3 mmHg.

Patient was extubated in the cath room.
Complication as previously noted: IC out with extravasation at Lt hand.

Procedure staffs:
1st operator: Dr. Drake
2nd operator: Dr. Neo
TEE operator: Dr. Roxanne
Instructor 1: Dr. Mao-Shin Lin (NTUH TPE)
Instructor 2: Dr. Shih-Hsien Sung (VGH TPE)

Figure 1.34: Risks of open heart surgery for MR STS score Fig ??

ful.

Management:

Eliquis 2.5 mg po bid until 1 day before operation.

Plan to continue in the coming morning.

Risks of open heart surgery for MR:

- Operative mortality: 10.8%
- Morbidity and Mortality: 24.6%
- Stroke: 3.85%
- Renal failure: 5.07%

(by ascdriskcalc.research.sts.org, accessed 2025-03-09)

Procedure steps:

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- Ultrasound-guided puncture to RFV, with pre-set of ProGlide.
- Transeptal puncture with SLO under TEE guidance, smoothly.
- Serial dilation and placement of 25 Fr SGC.
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- TEE operator: Dr. Roxanne
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- Instructor 2: Dr. Shih-Hsien Sung (VGH TPE)

1.13.3 Risk documentation- 醫師醫師醫師 operative risks

網址: <https://acsdriskcalc.research.sts.org/> 醫師醫師醫師 STS risk score. 醫師醫師醫師醫師醫師醫師醫師醫師醫師 (醫師醫師醫師) 醫師醫師醫師醫師 operative mortality 醫師 10.8%.

STS Short-term / Operative Risk Calculator
Adult Cardiac Surgery Database - All Procedures

Planned Surgery
Surgery Incidence: ☐ Isolated MVR
Surgical Priority: ☐ Front CAB surgery ☐ Back CAB surgery

Demographics
Sex: ☐ Male ☐ Female
Age (years): 64
Height (cm): 165
Weight (kg): 72
BMI (kg/m²): 26.1
Race: Asian
Payer / Insurance: Non-US Plan

Laboratory Values
Creatinine (mg/dL): 1.5
Hemoglobin (g/L): 35
WBC Count (10³/L): 8.4
Platelet Count (10³/L): 170000

Preoperative Medications
☐ ACE Inhibitors within 48 hrs
☐ QTc Interval within 5.0 ms
☐ Insulin within 48 hrs
☐ Steroids within 5 days
☐ ADP Inhibitors within 5 days

Risk Factors/Comorbidities
Diabetes: ☐ No ☐ Yes
Family Hx of CAD: ☐ No ☐ Yes
Hypertension: ☐ No ☐ Yes
Lung Disease: ☐ No ☐ Yes
Medication: ☐ No ☐ Yes
Unresponsive State: ☐ No ☐ Yes
Endocarditis: ☐ No ☐ Yes
Rxn Drug Use: ☐ No ☐ Yes
Alcohol Use: ☐ No ☐ Yes
Tobacco Use: ☐ No ☐ Yes
Pulmonary: ☐ No ☐ Yes
Chronic Lung Disease: ☐ No ☐ Yes
Recent Pneumonia: ☐ No ☐ Yes
Sleep Apnea: ☐ No ☐ Yes
Vascular: ☐ No ☐ Yes
Coronary Artery Disease: ☐ No ☐ Yes
Peripheral Artery Disease: ☐ No ☐ Yes
Right Carotid Sten. $\geq 80\%$: ☐ No ☐ Yes
Carotid Status: ☐ No ☐ Yes
NYHA Classification: ☐ Class I ☐ Class II ☐ Class III ☐ Class IV
Pulmonary Vascular Resistance: ☐ No ☐ Yes
Ejection Fraction (%): 55
Cardiogenic Shock: ☐ No ☐ Yes

Coronary Artery Disease
Prior: ☐ No ☐ Yes
Myocardial Infarction when: ☐ No ☐ Yes
No. of Diseased Vessels: 1
Valve Disease: ☐ No ☐ Yes
Aortic Disease: ☐ No ☐ Yes
Aortic Root Abscess: ☐ No ☐ Yes
Aortic Regurgitation: ☐ Mild ☐ Moderate ☐ Severe
Mitral Regurgitation: ☐ Mild ☐ Moderate ☐ Severe
Tricuspid Regurgitation: ☐ Mild ☐ Moderate ☐ Severe
Atrial Fibrillation: ☐ No ☐ Yes
Atrial Type: ☐ No ☐ Yes
Atrial Flutter: ☐ No ☐ Yes
V. Tach/V. Fib: ☐ No ☐ Yes
Sick Sinus Syn: ☐ No ☐ Yes
2nd Degree Block: ☐ No ☐ Yes
3rd Degree Block: ☐ No ☐ Yes
Previous Cardiac Interventions (Select all that apply): ☐ CABG ☐ Valve ☐ PCI ☐ Other

Simulated Patient Summary
Procedure Type: **Isolated MVR**

Postoperative Outcome	Estimate %
Operative Mortality	10.8%
Morbidity & Mortality	24.6%
Stroke	3.85%
Renal Failure	5.07%
Reoperation	7.52%
Prolonged Ventilation	13.5%
Deep Sterile Wound Infection	0.99%
Long Hospital Stay (>14 days)	18.2%
Short Hospital Stay (≤ 14 days)	6.86%

Clinical Summary
Planned Surgery: Isolated MVR, Elective, First cardiovascular surgery
Demographics: 64 year old, Asian, female, 5'4", 165cm, BMI 26.1 kg/m²
Insurance/Payer: Non-US Plan
Lab Values: Creatinine: 1.5 mg/dL, Hemoglobin: 35%, WBC Count: 8.4 x 10³/L, Platelet Count: 170000 cells/L
Substance Abuse: Never smoker
Risk Factors / Comorbidities: Hypertension, Family Hx of CAD
Vascular RF: Coronary Artery Disease CAB ≥ 80 days, Peripheral Artery Disease
Cardiac Status: Chronic heart failure, NYHA Class III, Ejection Fraction $\geq 55\%$
Coronary Artery Disease: No coronary symptoms
Valve Disease: Mitral Aortic Stenosis, Mitral Aortic Regurgitation
Atrial Fibrillation: Recent A-Fib, Persistent
PSP (Patient Summary) Copy

Figure 1.35: 醫師醫師醫師 operative mortality 醫師 10.8%. 醫師 Fig 醫師醫師 operative mortality: 10.8% 醫師醫師

2

TAVI in AS

2.1 Patient selection for TAVI in AS

TAVI in AS □□□□

1. symptomatic AS
2.

00000000: 000000000000AS000000/00000LV000000000 LVEF < 50%00

high gradient AS CW (continuous wave) ΔP (trans-valvular mean pressure gradient) ≥ 40 mmHg V_{max} (trans-valvular peak velocity) ≥ 4 m/s.

2.1.1 MDCT and fluoroscopy-guidance

2.2 Pre-TAVI evaluation by MDCT

MDCT 0000000000000000¹: ¹; and

- para-valvular leak
- coronary acute closure
- aortic annular rupture
- AV block leading to PPM implantation
- vascular complications

2.2.1 Acute coronary occlusion after TAVI

Figure 2.1: MDCT images showing the aortic annular plane and hinge points. The aortic annular plane is defined by the hinge points of the aortic valve. The distance between the hinge points is 29.1 mm x 21.6 mm. The area is 4.51 cm². The average diameter is 24.0 mm. The perimeter is 77.4 mm (5 mm smoothing). The coronary height is 14.6 mm. The LVOT is 181.2 ± 211 mm. The Agatston score is 22.5 ± 37.6. The P value is < 0.001. The LVOT is 20% of the Valsalva sinuses.

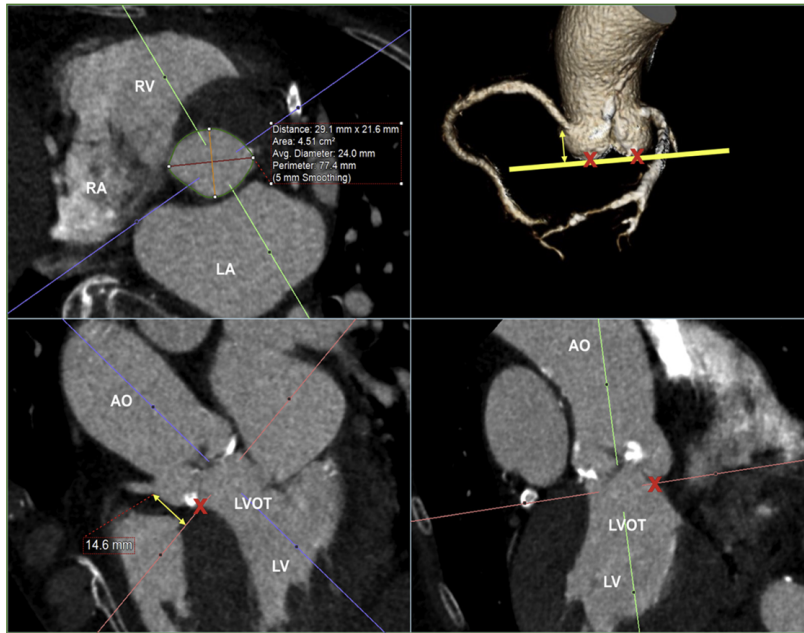


Figure 2.1: MDCT images showing the aortic annular plane and hinge points. The aortic annular plane is defined by the hinge points of the aortic valve. The distance between the hinge points is 29.1 mm x 21.6 mm. The area is 4.51 cm² (perimeter) 77.4 mm. Sagittal oblique view shows the coronary height (14.6 mm). The LVOT is 181.2 ± 211 mm. The Agatston score is 22.5 ± 37.6. The P value is < 0.001. The LVOT is 20% of the Valsalva sinuses.

^a [

2

coronary height $\geq 12\text{mm}$, SoV and STJ $\geq 28\text{ mm}$, coronary occlusion

2.2.2 Aortic annular rupture risk evaluation

AVA MDCT LVOT/Agatston 181.2 ± 211 22.5 ± 37.6 P < 0.001 LVOT Valsalva 20% ³

3

2.2.3 LVOT calcium grading

LVOT calcium grading definitions. Figure ??.

00	00
10	00 THV 000
11	00000for Self-expanding THV: 000 2/3 0000000
12	0000 THV0
13	000000000000000000 THV 00000000
14	000000000
15	00000

TAVI 手術のリスクを軽減するための新しい治療法として、Mainz 大学の TAVI 手術（conscious sedation）が注目されています。この手術は、患者が意識を失わずに手術を受けることができます。

[illegible]

2.5.1 Sizing issue

00000000 3D CT 0000000000THV0000000000

THV (transcatheter human valve) 3D CTA annular diameter perimeter (mm) SOV (sinus of Valsalva SOV) 3D CTA annular diameter perimeter (mm)

Table 2.3: Sizing issues for TAVI

Parameter	Clinical relevance
Annular diameters	Annular diameter at systole □□□□□□□□□□ THV □□□□□□□□□□ minimal annular diameter □□□□□□□□□□□□□□ average annular diameter □□□□□□□□□□□□□□
LVOT	□□□□□□□□ LVOT □□□□□□□□ LVOT or annulus rupture □□□
SoV, sinus of Valsalva	SoV diameter □□□□□□□□ 28 mm.

5/5/2019

Brecker/Medtronic, Lunderquist/Cook Medical 的 過渡點 (transition point) 位於 BAV 的 30 度角內，THV delivery system 的 BAV 位於 30 度角內，THV delivery system 的 BAV 位於 30 度角內。

2.5.4 Valve Implantation

annulus. (adjust) anatomy. capsule 及 cone, system cone 及 capsule Lunderquist/Cook 3-5 mm co-planar (aortic arch) 。

2.5.5 Self-expandable THV

1/3 1:1 90-130 度。 80% 。

2.5.6 balloon-expandable THV

Mainz balloon-expandable THV Sapien 3/Ultra. high implantation technique.⁴

CENTRAL ILLUSTRATION Clinical outcomes after TAVI according to prosthesis type and LVOT calcification.

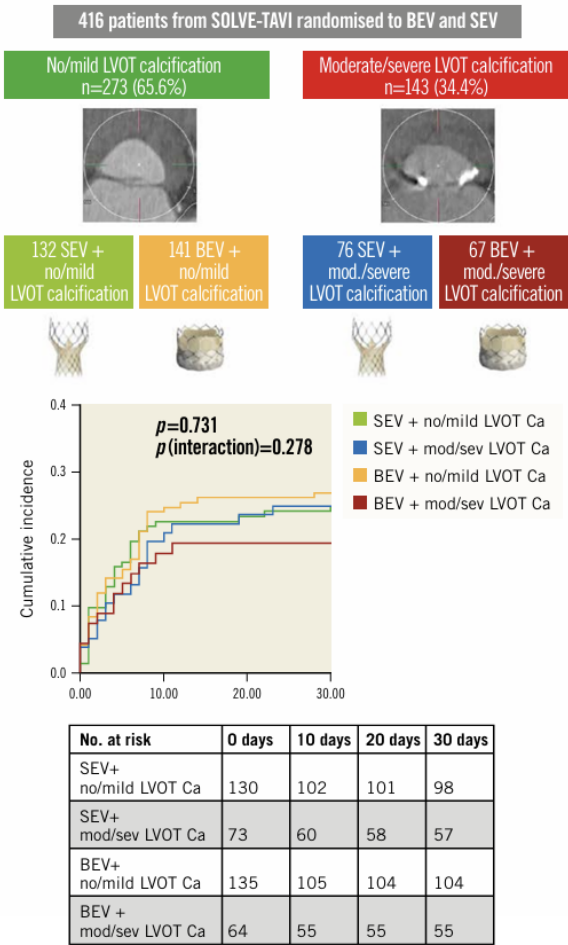


Figure 2.3: LVOT calcification was analyzed in 416 patients from the SOLVE-TAVI trial, which randomized patients to TAVI with either BEV or SEV and either general anesthesia or conscious sedation. There was no significant difference in the incidence of a composite endpoint of all-cause mortality, stroke, moderate or severe paravalvular regurgitation, permanent pacemaker implantation and annulus rupture. There was no interaction between prosthesis type and LVOT calcification. BEV: balloon-expandable valve; Ca: calcification; LVOT: left ventricular outflow tract; SEV: self-expanding valve; TAVI: transcatheter aortic valve implantation.^{4]}

^a [

2.5.7 Retrieval

2.6.2 Treatment protocol

Table 2.4: Device list for TAVI

□□	□□
□□□ (sheaths)	• 6 Fr sheaths x2 (□□□□ Rt CFA, □□□□ Rt radial artery). • 9 Fr □□□□ 6 Fr □□ Rt CFA. • TPM sheath: Rt CFV □□□□□ TPM sheath.
□□ (catheters)	• Pig-tail catheters, 6 Fr x2 • AL1 catheter, 6 Fr
□□ (wires)	• J-tip wire • Safair XS • Terumo GlideWire straight tip
□□□□□□□	• □□□□□□□□□□□□□□□□ • DI shock □□□□□□
□□□	Philips EPIC □□□□□ linear probe, sector probe □□□□□□□□ □□□□□□
□ □ □ □ (trancathter hu- man valve, THV)	• Evolute Fx+ • Sapien 3

2.7 Low-flow and low-gradient severe AS

Use dimensionless index □□□□:

$$\frac{VTI_{LVOT}}{VTI_{Ao}}$$

, 則

$$\frac{VTI_{LVOT}}{VTI_{Ao}} < 0.25$$

低壓、低梯度嚴重AS。

2.7.1 MDCT and aortic valve calcium score

利用 Doppler echocardiography 評估瓣膜 (AS) 嚴重程度與 MDCT 評估瓣膜 (aortic valve calcium, AVC) 嚴重程度之比較研究顯示，MDCT 評估瓣膜嚴重程度與 Doppler echocardiography 評估瓣膜嚴重程度之相關性較低。

AVC 與 MDCT 評估瓣膜嚴重程度之 modified Agatston 評估瓣膜嚴重程度之 130 Hounsfield 單位評估瓣膜嚴重程度 AS 評估瓣膜嚴重程度 AVC 評估瓣膜嚴重程度之相關性較低。

評估瓣膜嚴重程度 AS 評估瓣膜嚴重程度之 modified Agatston 評估瓣膜嚴重程度之 130 Hounsfield 單位評估瓣膜嚴重程度 AS 評估瓣膜嚴重程度 AVC 評估瓣膜嚴重程度之 AS 評估瓣膜嚴重程度之 paradoxical low-flow, low-gradient (評估瓣膜嚴重程度) AS 評估瓣膜嚴重程度 normal-flow, low gradient (評估瓣膜嚴重程度) AS 評估瓣膜嚴重程度 DSE 評估瓣膜嚴重程度 AVC 評估瓣膜嚴重程度 AS。

評估 AS 評估瓣膜嚴重程度 AVC 評估瓣膜嚴重程度 2000 Agatston 評估瓣膜嚴重程度 1200 Agatston 評估瓣膜嚴重程度。

評估瓣膜嚴重程度 severe AS 評估瓣膜嚴重程度 AVC score 評估瓣膜嚴重程度。

2.8 Choice between different THV in TAVI procedures

評估瓣膜嚴重程度 Mainz 評估瓣膜嚴重程度 heart team 評估瓣膜嚴重程度。

1. Durability: 評估瓣膜嚴重程度 80-90 評估瓣膜嚴重程度 THV 評估瓣膜嚴重程度 durability, 評估瓣膜嚴重程度 procedure 評估瓣膜嚴重程度 60 評估瓣膜嚴重程度 20 評估瓣膜嚴重程度 survival, 評估瓣膜嚴重程度 durability 評估瓣膜嚴重程度 THV。評估瓣膜嚴重程度: 2024-11 評估瓣膜嚴重程度: durability for Sapine 3 is better then Evolute. 評估瓣膜嚴重程度 2024-11-12 Case 22 評估瓣膜嚴重程度 Sapien 3 (評估瓣膜嚴重程度 62 評估瓣膜嚴重程度, Dx: bisucpid AS)。
2. 評估瓣膜嚴重程度 (LVOT calcificatoin) 評估瓣膜嚴重程度 (annular rupture) 評估瓣膜嚴重程度 self-expandable THV。
3. 評估瓣膜嚴重程度 (short membranous septum) 評估瓣膜嚴重程度 AV block 評估瓣膜嚴重程度 balloon-expandable THV 評估瓣膜嚴重程度 His bundle 評估瓣膜嚴重程度。

4. 完全性 AV block 膜性室间隔缺损 完全性左束支传导阻滞
self-expandable THV, 经导管 后扩张 完全性 AV block.

2.9 Post-TAVI imaging follow-up

经导管 TAVI 术后影像学随访 MDCT 和 echo.⁵

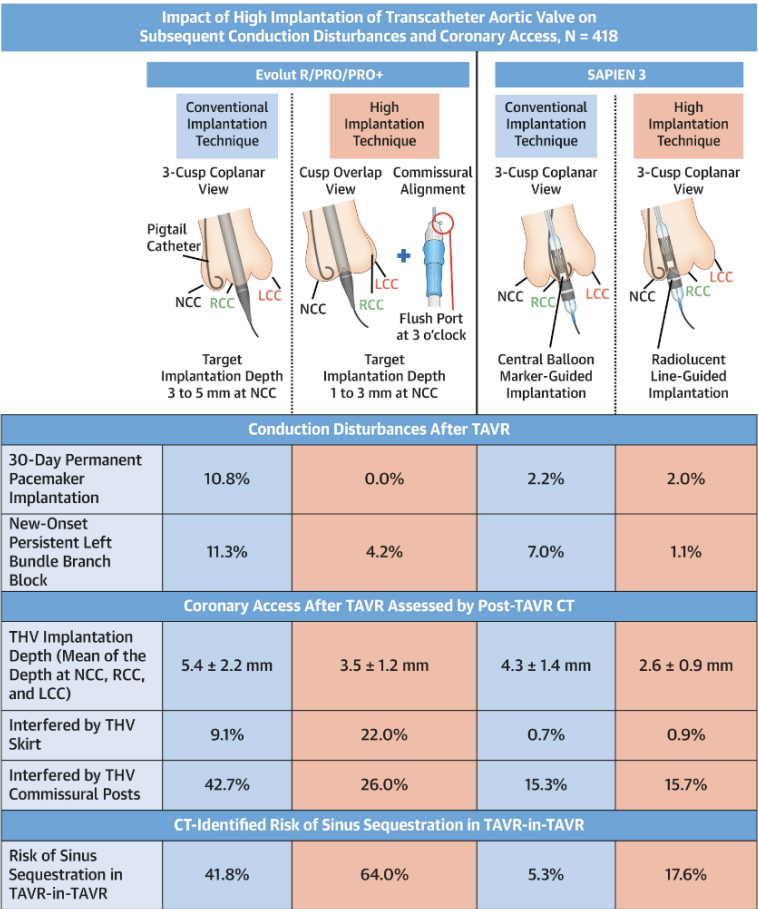


Figure 2.5: 经导管 (high implantation) 经导管 TAVI 术后影像学随访 MDCT 和 echo.⁵

Ochiai T, et al. J Am Coll Cardiol Interv. 2023;16(10):1192-1204.

经导管 TAVI 术后影像学随访 MDCT 和 echo.⁵

Table 2.5: TAVI 檢查項目

項目	說明
MDCT after TAVI	• THV 完整性 (integrity) • 位置 (position) • 貼附 (apposition) • 圓形 (circularity) • 擴張 (expansion) • 旁瓣漏 (para-valvular leak) • 併發症 (complications): 血栓 (thrombosis) 瓣膜病 (pannus formation)
Echocardiography after TAVI	• 檢查 2D 瓣膜關閉情況 • 檢查瓣膜關閉情況 EOA 瓣膜關閉情況 • 檢查瓣膜關閉情況 • 檢查瓣膜關閉情況
THV stenosis	THV stenosis is most often associated with thickening and reduced mobility of the leaflets due to thrombosis, fibrosis, and/or calcification
THV leaflet thickening	Valve leaflet thickening is generally defined as a valve leaflet with thickness ≥ 2 mm.

鄧國權

TAVI in AR

1

Trilogy 株式会社 JenaValve Technology 株式会社
 日本 大阪府 大阪市 北区 西成区 1-1-1
 〒545-0051

□ 2018 □ 2022 □□□□□□ 346 □□□□□□□□ 180 □□□□□□□□□□ 75.5 □□ □□□ 47%□□□□□□□□ 95%. 30 □□□□□ 2%, □□□□□ 1%. □□□□□□□□□□□□ □□□□ 27%□p<0.0001□□24% □□□□□□□□□ permanent pacemaker. 1 □□ □□□□ 7.8%□p<0.0001□□

2

49

Table 3.1: ALIGN-AR □□□□□□□□

[illegible]

Table 3.2: Detailed ALIGN-AR trial patient selection criteria

Type	Criteria
Inclusion Criteria	
1	Adult subjects with severe AR (Grade ≥ 3) based on ASE guidelines: • Jet width $\geq 65\%$ of LVOT. • Vena contracta width > 6 mm. • Holodiastolic flow reversal in the proximal abdominal/descending aorta. • Jet deceleration rate/Pressure half-time < 200 ms. • Regurgitant volume and fraction consistent with Grade 3 or 4 AR thresholds.
2	Patient symptomatic according to NYHA functional class II or higher.
3	High surgical risk for SAVR as documented by Heart Team, with agreement that SAVR can be used as a “bailout”.
4	Suitable anatomy to accommodate the JenaValve Trilogy™ Heart Valve System.
5	Written informed consent provided by the patient or legal representative.

Type	Criteria (continued)
6	Willingness to comply with all post-procedure follow-up requirements.
Exclusion Criteria	
1	Congenital uni- or bicuspid aortic valve morphology.
2	Previous prosthetic aortic valve implantation.
3	Mitral regurgitation > moderate.
4	Clinically significant CAD requiring revascularization within 30 days before procedure or planned revascularization within 12 months post-procedure.
5	Echocardiographic evidence of left ventricular thrombus.
6	Endocarditis within the last 180 days.
7	Hypertrophic cardiomyopathy with or without obstruction.
8	Severe pulmonary hypertension (systolic PA pressure > 80 mmHg).
9	Severe RV dysfunction as assessed clinically and via echocardiography.
10	Severely reduced LVEF (< 25%).
11	Aortic annular perimeter-derived diameter < 21.0 mm or > 28.6 mm, or perimeter < 66 mm or > 90 mm.
12	Aortic annulus angulation > 70°.
13	Straight length of ascending aorta < 55 mm.
14	Significant disease of ascending aorta (e.g., aneurysm $\geq$ 50 mm or severe atheroma).
15	Urgent or emergent TAVI required.
16	Cardiogenic shock or hemodynamic instability requiring inotropic support or ventricular assist device.
17	Myocardial infarction < 30 days before procedure.
18	Cerebrovascular event (TIA or stroke) < 180 days before procedure.
19	Severe renal insufficiency (GFR < 30 ml/min) or recent renal replacement therapy (< 180 days).

Type	Criteria (continued)
20	Blood dyscrasias (e.g., WBC < 3000/mm ³ , platelets < 90,000/ μ l, or anemia [Men: Hgb < 8.1 g/dl; Women: Hgb < 7.4 g/dl]).
21	Active peptic ulcer or gastrointestinal bleeding < 90 days before procedure.
22	Known hypersensitivity to aspirin, heparin, ticlopidine, clopidogrel, nitinol, tantalum, or contrast agents.
23	Contraindication to intraoperative transesophageal echocardiography or MDCT scan.
24	Estimated life expectancy < 24 months.
25	Enrollment in another investigational study.
26	Medical, social, or psychological conditions that preclude participation as per investigator.
27	Severe dementia impacting consent or compliance with follow-up.
28	Inability to comply with follow-up requirements.

3.2 ALIGN-AR trial with focus on Trilogy valve

Table 3.3: Trilogy Valve in ALIGN-AR trial

Item	Criteria
Item	
Item	
Item	- Item - Item - Item Locators
Item	Item
Item	Item
Item	

4

11

125 Mainz Heart Valve Center: Dr. Ralph Stephan von Bardeleben, Dr. Ruf Tobias, Dr. Theresa, Dr. Julian Dr. Kabir (from India) and Dr. Julian (from Colombia)

□□□□□□□□□□□□□□□□□□ (Figure ??)□

[illegible]

overleaf (Latex). Latex Tufte-Book
template ChatGPT prompt : " "
" Latex package:
longtable ChatGPT Latex (prompt : longtable
latex longtable)

聯絡電話: H11135@hch.gov.tw. ¹

1

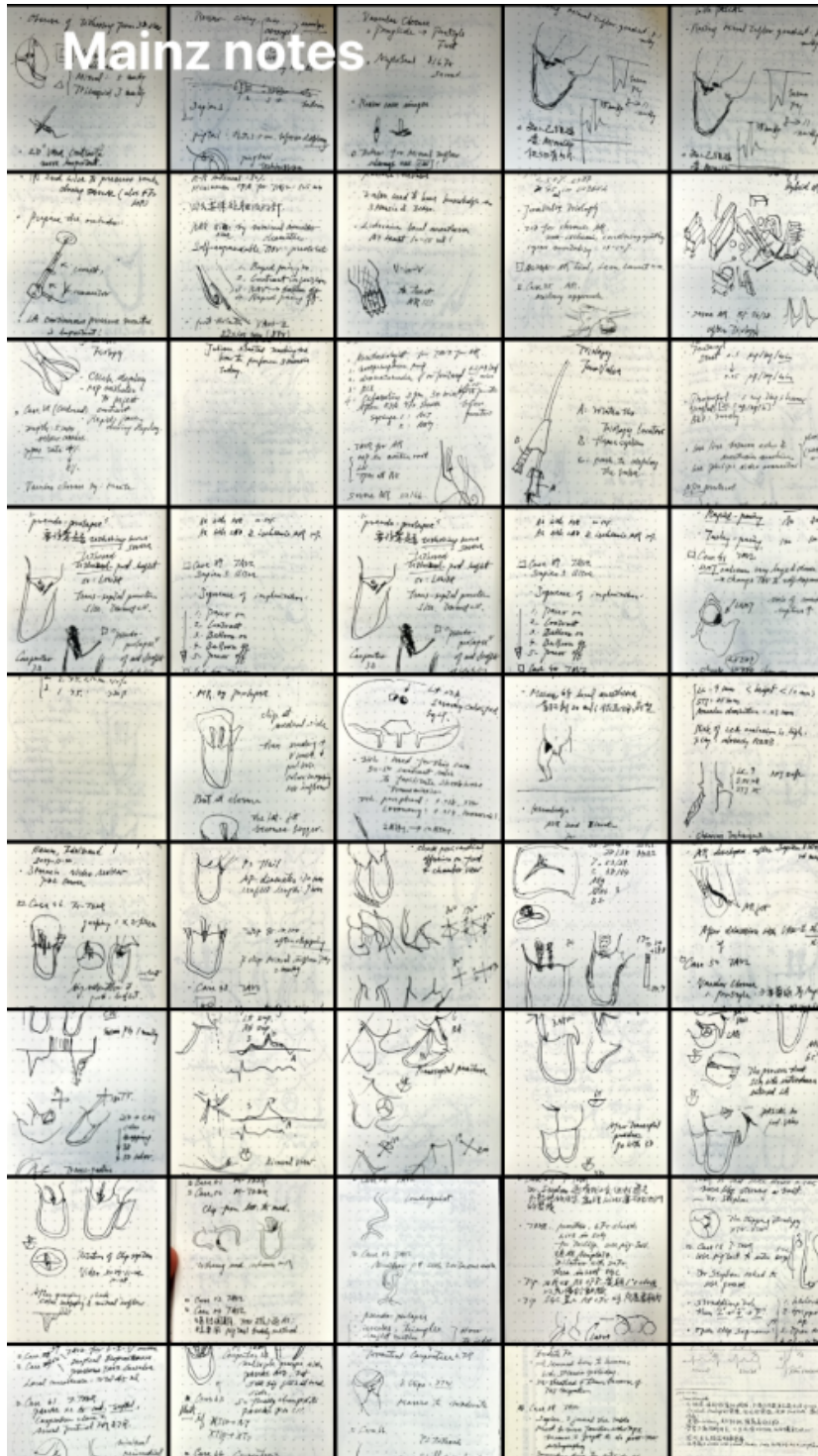


Figure 4.1: □ Mainz Hybrid OR □□□□□□ □□□□□□□□